

Computational Finance (WI4151) — Theory

Cheat-Sheet

One block per lecture: core idea, key results, the line of reasoning. No derivations.

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Notation. $x = \ln S$ or $\ln(S/K)$ | $\varphi_X(u) = \mathbb{E}[e^{iuX}]$ (cf) | $N(\cdot)$ normal CDF, $\phi(\cdot)$ normal pdf | $[a, b]$
 COS range | $\sum_{k=0}^{N-1'}$ first term halved | c_1, c_2, c_4 cumulants | $\omega_k = \frac{k\pi}{b-a}$ | r, T, σ, K as usual.

The single through-line. Every price is a discounted risk-neutral expectation

$$V = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\text{payoff}] = e^{-rT} \int_{\mathbb{R}} V(y, T) f(y | x_0) dy.$$

If the *density* f is known (GBM) use closed forms / PDEs / trees; if only the *cf* φ is known (Heston, jumps, AJD) use **COS**. Early exercise (L9) and discrete barriers (L10) reuse COS as a backward recursion; high-dimensional or averaging payoffs (L11) fall back to Monte Carlo (L4). **COS (L6, L10) is the centre of gravity of the course.**

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1 Cluster 1 — Foundations, lattice, PDE (L1–L3, L5)

Lecture 01 — Asset classes, no-arbitrage, the Wiener process

Core idea. The pricing primitives: time value of money, no-arbitrage replication, and the stochastic model of S .

- **Discounting.** $dM = rM dt \Rightarrow M(t) = e^{rt}$; one euro at T is worth $e^{-r(T-t)}$. Covered interest-rate parity $(1 + i_s) = (F_t/S_t)(1 + i_c)$ by static replication; the “FX basis” is the empirical correction restoring equality.
- **Payoffs & decomposition.** Call $(S_T - K)^+$, put $(K - S_T)^+$; value = intrinsic + time value; both payoffs kink at $S = K$ (the kink is why a *direct* Fourier expansion of the payoff rings; COS expands the smooth density instead).
- **One-step replication.** Hedge ratio $\Delta = \frac{V_u - V_d}{S_u - S_d}$; the no-arbitrage price equals $\mathbb{E}^Q[\text{payoff}]$ and physical probabilities never enter. Risk-neutral $q = \frac{S_0 - S_d}{S_u - S_d}$ (with $r = 0$) makes $\mathbb{E}^Q[S_T] = S_0$. A perfectly hedged writer makes zero in both states ($\Pi_u = \Pi_d = 0$): the fair price is exactly the replication cost.
- **Bounds & parity.** $C \geq (S - Ke^{-r(T-t)})^+ > S - K$ for $r > 0 \Rightarrow$ an American call on a non-dividend stock is *never* exercised early (= European). Put–call parity $C + Ke^{-r(T-t)} = P + S$ is model-free (static $S_T - K$ replication).
- **Model.** GBM $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$, so $\ln(S_T/S_0) \sim N((\mu - \frac{1}{2}\sigma^2)T, \sigma^2 T)$, $\mathbb{E}[S_t] = S_0 e^{\mu t}$. Wiener: $W_0 = 0$, independent stationary increments, $W_t \sim N(0, t)$, continuous paths; $\mathbb{E}[dW] = 0$, $(dW)^2 = dt$, $[W]_t = t$. A zoo of SDEs: ABM (can go negative), GBM (multiplicative, positive), OU (mean-reverting) — OU is the template for the Heston/CIR variance.

Reasoning line. Price = cost of the replicating hedge $\Rightarrow$ risk-neutral expectation; $(dW)^2 = dt$ forces Itô calculus; GBM is the baseline every later model perturbs.

Likely question: Why do the physical probabilities p never enter the one-step price?

Lecture 02 — Itô, the Black–Scholes PDE, risk-neutral valuation, Feynman–Kac

Core idea. Three equivalent views of the European price: a hedging PDE, a $\mathbb{Q}$ -expectation, and Feynman–Kac between them.

- **Itô.** $dg = (g_t + \mu g_x + \frac{1}{2}\sigma^2 g_{xx})dt + \sigma g_x dW$; second-order term survives because $(dW)^2 = dt$. On $g = \ln S$: $d \ln S = (\mu - \frac{1}{2}\sigma^2)dt + \sigma dW$ (the $-\frac{1}{2}\sigma^2$ is the convexity correction that keeps $\mathbb{E}[S_t] = S_0 e^{\mu t}$).
- **BS PDE.** $V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0$; delta-hedge $\Delta = V_S$ cancels dW , the riskless portfolio earns r , and the drift μ drops out. Solved backward from $V(S, T) = \text{payoff}$.
- **Risk-neutral measure.** Under $\mathbb{Q}$: $\mu \rightarrow r$, σ unchanged (equivalence preserves quadratic variation); $e^{-rt}V$ is a $\mathbb{Q}$ -martingale. Feynman–Kac: $V(S, t) = e^{-r(T-t)}\mathbb{E}^Q[\text{payoff} | S_t]$.

European call: $C = S_0 N(d_1) - Ke^{-rT} N(d_2)$, $d_{1,2} = \frac{\ln(S_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, $d_2 = d_1 - \sigma\sqrt{T}$, $N(d_2) = \mathbb{Q}(S_T > K)$.

Greeks (call): $\Delta = N(d_1)$, $\Gamma = \frac{\phi(d_1)}{S_0\sigma\sqrt{T}}$, $\text{Vega} = S_0\sqrt{T}\phi(d_1)$, $\Theta = -\frac{S_0\phi(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)$, $\rho = KTe^{-rT}N(d_2)$.

The BS PDE re-read at S_0 : $\Theta + rS_0\Delta + \frac{1}{2}\sigma^2 S_0^2 \Gamma = rC$ (an identity linking the Greeks).

Reasoning line. The hedge removes dW so the portfolio earns r ; μ cancels, which is precisely why the real-world drift is absent from every option price.

Likely question: Where exactly in the delta-hedging derivation is $\Delta = V_S$ forced, and where does μ cancel?

Lecture 03 — Implied volatility

Core idea. Invert the BS price for its one unknown σ ; the IV surface is how the market's view is read off prices.

- Four knowns (S, K, T, r) , one unknown σ : a monotone root-find, not a formula.
- **Existence/uniqueness.** $\text{Vega} = S\sqrt{T}\phi(d_1) > 0 \Rightarrow$ price strictly increasing in $\sigma \Rightarrow$ a unique IV, provided the quote sits inside the no-arbitrage interval $[(S - Ke^{-rT})^+, S]$ (call).
- **Inversion.** Bisection: linear, bracket-robust, $\sim \log_2(\text{width}/\text{tol})$ steps. Newton: quadratic but the step $\sigma - f/\text{vega}$ blows up where $\text{vega} \rightarrow 0$ (deep OTM / short T); globalise by seeding at the price-curve inflection point in σ (monotone convergence from there).
- Put and call at the same (K, T) share one IV (parity). Surface: smile/skew across K , term structure across T ; the SPX QQ-plot (fat tails) is the empirical reason BS fails.
- **Three notions:** *implied* (a quoting convention, one number per quote), *local* $\sigma(S, t)$ (a deterministic function fitting the whole surface), *stochastic* (variance has its own SDE — Heston).

Reasoning line. Monotonicity ($\text{vega} > 0$) guarantees a unique root; a non-flat surface is the empirical proof that constant- σ BS is wrong — this is what motivates Heston (L7) and jumps (L8).

Exam trap. Newton's $1/\text{vega}$ step is unsafe where $\text{vega} \rightarrow 0$; bisection is the safe fallback.

Lecture 05 — Binomial trees and finite differences

Core idea. Two lattice/grid pricers — and the fact that a binomial tree *is* an explicit FD scheme.

- **CRR.** u, d, p from matching the mean and variance of the tree log-return to BS; the Itô-corrected drift $r - \frac{1}{2}\sigma^2$ (not r) sits inside u, d . Price by risk-neutral backward induction $V_m^n = e^{-r\Delta t}(pV_{m+1}^{n+1} + (1-p)V_m^{n+1})$. Convergence $O(1/M)$, *non-monotone* (sign-oscillating as the strike straddles nodes).
- **FD on the heat equation** $u_t = u_{xx}$. FTCS (explicit), BTCS (implicit), Crank–Nicolson ($\frac{1}{2}\text{FTCS} + \frac{1}{2}\text{BTCS}$). Local truncation: FTCS/BTCS $O(k + h^2)$, CN $O(k^2 + h^2)$.
- **Von Neumann.** $\xi_{\text{FTCS}} = 1 - 4\nu \sin^2(\beta h/2)$ with $\nu = k/h^2$; $|\xi| \leq 1 \Rightarrow \nu \leq \frac{1}{2}$ (CFL: $k \leq h^2/2$). $\xi_{\text{BTCS}} = (1 + 4\nu \sin^2(\beta h/2))^{-1}$, unconditionally stable.
- **Lax:** consistency + stability $\Rightarrow$ convergence (consistency alone is not enough).
- **BS $\rightarrow$ heat.** $x = \ln S$ (constant coefficients), reverse time $\tau = T - t$, $V = e^{-r\tau}W$. Binomial = FTCS with the coupling $h^2 = \sigma^2 k$ collapsing the centre node into the two-point recursion.

Reasoning line. Explicit schemes are cheap but the CFL bound forces $k = O(h^2)$; working in $x = \ln S$ gives constant coefficients and tames the large- S diffusion weight $\propto j^2$ — the same domain-truncation lesson as the COS range $[a, b]$.

Likely question: In what precise sense is the binomial tree an explicit FD scheme “specialised to a single time-zero value”?

2 Cluster 2 — Monte Carlo (L4)

Lecture 04 — MC integration, SDE simulation, variance reduction

Core idea. An unbiased sampling estimator with *dimension-independent* $O(N^{-1/2})$ error — plus the tools that shrink its constant.

- **Estimator.** $\hat{I} = \frac{1}{N} \sum_i g(X_i)$, unbiased, $\text{Var}(\hat{I}) = \sigma^2/N$, standard error $s_N/\sqrt{N}$ (use the $N - 1$ sample variance), CLT interval $\hat{I} \pm z_{\alpha/2}s_N/\sqrt{N}$. The rate is d -independent, unlike tensor quadrature $O(N^{-k/d})$.

- **QMC.** Low-discrepancy points $\Rightarrow$ error $\leq D_N^*$ (star discrepancy) $\times V(g)$ (Koksma–Hlawka), giving $\sim (\log N)^d/N$, near $O(N^{-1})$. Degrades in high effective dimension; randomised QMC restores an error estimate.
- **SDE schemes.** Euler–Maruyama $X_{n+1} = X_n + a h + b \Delta W$ (strong $\frac{1}{2}$, weak 1); Milstein adds $\frac{1}{2}bb'[(\Delta W)^2 - h]$ from the Itô–Taylor expansion (strong 1). Strong = pathwise error, weak = error in distribution / expectations.
- **Variance reduction.** Antithetic ($\pm Z$; needs monotone payoff so the pair is negatively correlated); control variate (optimal $c^* = \text{Cov}(X, Y)/\text{Var}(Y)$, residual variance factor $1 - \rho^2$); importance sampling (likelihood-ratio reweight / Girsanov drift shift; beware blow-up to infinite variance).

Reasoning line. MC wins in high dimension because the rate is d -free; it is the slow $O(N^{-1/2})$ benchmark that COS and FFT beat for smooth, low-dimensional European problems.

Likely question: Derive the Milstein correction term and state the strong/weak orders of Euler vs Milstein.

3 Cluster 3 — Fourier / COS (L6, L10) — PRIORITY

Lecture 06 — The Fourier family and the COS method

Core idea. Compute $e^{-rT}\mathbb{E}[\text{payoff}]$ using *only* the cf, by expanding the density in a cosine series whose coefficients are read straight off φ .

- **Fourier family.** cf $\varphi_X(u) = \int e^{iux} f(x) dx$ is the Fourier dual of f ; inversion $f(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$. The cf is known in closed form for a long list of models even when f is not. DFT/FFT is $O(N \log N)$; the half-range *cosine* series (even extension) underlies COS.
- **Carr–Madan (precursor).** Undamped call is not L^1 in log-strike k ; damp $c_T(k) = e^{\alpha k} C_T(k)$, then $\psi_T(v) = \frac{e^{-rT} \varphi(v - (\alpha + 1)i)}{\alpha^2 + \alpha - v^2 + i(2\alpha + 1)v}$, invert by FFT. Weaknesses COS removes: a numerical integration and a free α .
- **Density recovery (engine).** On $[a, b]$, $f(y) \approx \sum_{k=0}^{N-1'} A_k \cos(\omega_k(y - a))$ with the cf-sampled coefficient

$$A_k \approx F_k = \frac{2}{b-a} \operatorname{Re} \left\{ \varphi(\omega_k) e^{-i\omega_k a} \right\}, \quad \omega_k = \frac{k\pi}{b-a}$$

(the *only* approximation is extending the integral to $\mathbb{R}$, i.e. range truncation).

European COS price: $P \approx e^{-rT} \sum_{k=0}^{N-1'} \operatorname{Re} \left\{ \varphi_{X_T}(\omega_k; x_0) e^{-i\omega_k a} \right\} V_k$.

Payoff coefficients ($y = \ln(S_T/K)$): call $V_k = \frac{2}{b-a} K [\chi_k(0, b) - \psi_k(0, b)]$, put $V_k = \frac{2}{b-a} K [\psi_k(a, 0) - \chi_k(a, 0)]$, with the elementary closed forms

$$\chi_k(c, d) = \frac{1}{1 + \omega_k^2} \left[\cos(\omega_k(d-a))e^d - \cos(\omega_k(c-a))e^c + \omega_k \sin(\omega_k(d-a))e^d - \omega_k \sin(\omega_k(c-a))e^c \right],$$

$$\psi_k(c, d) = \begin{cases} [\sin(\omega_k(d-a)) - \sin(\omega_k(c-a))]/\omega_k, & k \neq 0, \\ d - c, & k = 0. \end{cases}$$

Cumulant range: $[a, b] = [c_1 - L\sqrt{c_2 + \sqrt{c_4}}, c_1 + L\sqrt{c_2 + \sqrt{c_4}}]$, $L \approx 8\text{--}12$. BS: $c_1 = \ln \frac{S_0}{K} + (r - \frac{1}{2}\sigma^2)T$, $c_2 = \sigma^2 T$, $c_4 = 0$.

- **Cost & convergence.** $O(N)$ per price, *no FFT, no quadrature*; exponential convergence for smooth (analytic-strip) densities, saturating at the range-truncation floor. Three errors: range

truncation (fixed floor), series truncation (set by smoothness $\Rightarrow A_k$ decay), cf/model error. Algebraic only when f has finite smoothness ($A_k = O(k^{-(\beta+1)})$), e.g. VG with small ν .

- **Why cosine, not complex Fourier:** (i) even extension $\Rightarrow$ real coefficients; (ii) A_k is literally $\text{Re}\{\varphi\}$ at the grid frequency; (iii) V_k closes in χ_k, ψ_k .
- **Greeks free:** S_0 enters only through φ (for Lévy/BS $\varphi(u; x_0) = \varphi_{\text{lvl}}(u)e^{iu x_0}$), so Δ, Vega are termwise derivatives of the same sum; one set of $\{V_k\}$ prices the whole grid.

Reasoning line. The cosine basis pairs the cf-derived density coefficients against closed-form payoff coefficients (a Parseval-style product); the only errors you control are the range $[a, b]$ and the term count N .

Short-maturity range collapse (Assignment 2 bug). As $T \rightarrow 0$, $c_2 = \sigma^2 T \rightarrow 0$, so the half-width $L\sqrt{c_2 + \sqrt{c_4}} \rightarrow 0$ and $[a, b]$ collapses toward c_1 (Gaussian has $c_4 = 0$, no floor). *Signature:* error grows as $T \rightarrow 0$ and does *not* improve with larger N — the tell-tale of a *range* error (a fixed floor), not a *series* error (curable by more terms). *Fix:* raise L at short T , floor the half-width / retain the $\sqrt{c_4}$ term, and construct $[a, b]$ to cover the whole spot/strike grid, not a single point.

Lecture 10 — Barrier options

Core idea. Payoff $\times$ survival indicator. Analytic under GBM via the reflection principle; for discrete monitoring a COS *backward recursion* that — unlike European COS — does use the FFT.

- **Definitions.** Eight types (up/down $\times$ in/out $\times$ call/put). In-out parity $C_{\text{in}} + C_{\text{out}} = \text{vanilla}$ (price the easier leg, subtract; free correctness check).
- **Two formulations.** Expectation $V = e^{-r(T-t)} \mathbb{E}_x[(\alpha(e^{X_T} - K))^+ \mathbf{1}_{\{\tau_B > T\}}]$ ($\alpha = \pm 1$); localized BS PDE with *absorbing* (Dirichlet) boundaries $V(a, t) = V(b, t) = 0$ at the knock-out level — the barrier enters through the boundary, not the equation.
- **Reflection principle (GBM).** $\mathbb{P}(\tau_a \leq T) = 2\mathbb{P}(W_T \geq a) \Rightarrow$ running-max density $p_{\Xi}(a) = \sqrt{2/\pi T} e^{-a^2/2T} \mathbf{1}_{a \geq 0}$; joint density $p_{\Xi, W_T}(a, b) = \sqrt{2/\pi} \frac{2a-b}{T^{3/2}} e^{-(2a-b)^2/2T} \mathbf{1}_{a \geq \max(b, 0)}$. This $p(\xi, y)$ integrates against the payoff to give closed-form barrier prices (drifted case by Girsanov).
- **PDE routes.** Method of images $U(S, t) = S^{2\alpha} V(B^2/S, t)$, $\alpha = \frac{1}{2}(1 - 2r/\sigma^2)$, e.g. $C_{\text{down-out}} = C_{\text{van}}(S, K) - (S/B)^{2\alpha} C_{\text{van}}(B^2/S, K)$; or a sine-series solution of the reduced heat equation (homogeneous Dirichlet $\Rightarrow$ sine basis).

Discrete COS (backward recursion). For $m = M, \dots, 2$: $\hat{c}(x, t_{m-1}) = e^{-r\Delta t} \sum_k' F_k(x) V_k(t_m)$, $F_k(x) = \text{Re}\{\varphi(\omega_k; x) e^{-i\omega_k a}\}$. Knock-out overwrites $v = R$ (rebate) above the barrier $h = \ln(H/K)$, splitting V_k into a payoff piece $\Psi_k(h, b)$ and a continuation piece $\hat{C}_k$ built from the previous $\{V_j\}$ through a coupling $M_{k,j}$ that is **Hankel + Toeplitz** $\Rightarrow$ each step is a convolution by **FFT**. Total cost $O((M-1)N \log_2 N)$.

Reasoning line. The Markov property factorises the survival product $\prod_m \mathbf{1}_{\{x_m < h\}}$ into one-step transitions, turning the price into a backward recursion of European-COS steps. **Key contrast:** European COS needs no FFT, but the discrete barrier must propagate *all* $\{V_k\}$ each step — a structured matrix–vector product — so the FFT returns. Continuous monitoring is the $M \rightarrow \infty$ limit (extrapolate in Δt); Heston barriers go 2-D (1-D COS in log-stock + integration over variance, or full 2-D COS).

4 Cluster 4 — Models (L7, L8)

Lecture 07 — Heston stochastic volatility

Core idea. Give variance its own mean-reverting Brownian shock; price by plugging the affine cf into COS.

- **Dynamics.** $dS = rS dt + \sqrt{v} S dW_1$, $dv = \kappa(\bar{v} - v) dt + \eta\sqrt{v} dW_2$, $d\langle W_1, W_2 \rangle = \rho dt$. Variance is CIR; $v_T \mid v_t$ is noncentral χ^2 .
- **Feller.** $2\kappa\bar{v} \geq \eta^2$ keeps $v > 0$ a.s. (else v touches 0 and the density/cf misbehave); derived from the noncentral- χ^2 density not blowing up at the origin.
- **Affine.** Not affine in (S, v) but *is* in $(\ln S, v)$, so the log-price cf is exponential-affine, from **Riccati ODEs**:

$$\varphi(u, \tau) = \exp\left(iu(\ln S_0 + r\tau) + \frac{v_0}{\eta^2} \frac{1 - e^{-D\tau}}{1 - Ge^{-D\tau}} (\kappa - i\rho\eta u - D) + \frac{\kappa\bar{v}}{\eta^2} [(\kappa - i\rho\eta u - D)\tau - 2 \ln \frac{1 - Ge^{-D\tau}}{1 - G}]\right),$$

$$D = \sqrt{(\kappa - i\rho\eta u)^2 + \eta^2(u^2 + iu)}, \quad G = \frac{\kappa - i\rho\eta u - D}{\kappa - i\rho\eta u + D}.$$

- **Simulation.** Euler can make $v < 0$ (full-truncation / reflection / log- v fixes, each biased); Broadie–Kaya *exact* via noncentral- χ^2 sampling of v + the cf of the integrated variance (generator / time-change / Girsanov machinery).
- **Smile intuition.** η controls curvature (smile), ρ the skew sign, κ the term-structure speed, $v_0/\bar{v}$ the level/long-run.

Reasoning line. Heston is the headline COS use case: one analytic cf prices the whole smile in a single $O(N)$ sum.

Exam trap. The “little Heston trap”: the complex log / $\sqrt{\cdot}$ branch cut destabilises the cf at long maturities — use the stable grouping (the form above). Negative ρ and small c_2 worsen the short-maturity range trap.

Lecture 08 — Jumps and the affine jump-diffusion framework

Core idea. Add discontinuities; the cf still closes in elementary form, so COS still applies unchanged.

- **Poisson.** $\mathbb{P}(N_t = k) = e^{-\lambda t} (\lambda t)^k / k!$, mean = var = λt ; compensated $N_t - \lambda t$ is a martingale; intensity may be time-dependent ($\Lambda = \int \lambda$) or stochastic (Cox).
- **Compound Poisson cf.** $Q_t = \sum_{i=1}^{N_t} Y_i$ has $\varphi(u) = \exp(t\lambda(\hat{\nu}(u) - 1))$ (the Lévy–Khintchine jump exponent), by conditioning on N_t . Adding a diffusion gives the full jump-diffusion log-price cf.
- **Merton SDE.** $dS/S_{t-} = (r - \lambda\mathbb{E}[J])dt + \sigma dW + (J - 1)dN$; drift correction $\mu = r - \lambda\mathbb{E}[J]$ (compensator) keeps $e^{-rt}S_t$ a $\mathbb{Q}$ -martingale. Value solves a non-local **PIDE**.
- **Jump-size laws.** Merton (lognormal jumps): cumulants of $\ln S_T$ are $c_1 = (r - \frac{1}{2}\sigma^2 - \lambda\kappa)T + \lambda T\mu_J$, $c_2 = T(\sigma^2 + \lambda(\mu_J^2 + \sigma_J^2))$, $c_4 = \lambda T(\mu_J^4 + 6\mu_J^2\sigma_J^2 + 3\sigma_J^4)$. Kou (double-exponential): $\hat{\nu}(u) = \frac{p\eta_1}{\eta_1 - iu} + \frac{(1-p)\eta_2}{\eta_2 + iu}$, needs $\eta_1 > 1$.
- **AJD (Duffie–Pan–Singleton).** cf = $\exp(\alpha(\tau) + \beta(\tau) \cdot x)$ from Riccati ODEs with the jump-transform term $(\theta(\beta) - 1)$; BS is the no-jump case.

Reasoning line. The compensator / $(\theta - 1)$ motif recurs (compensated Poisson, drift correction, AJD ODE); the only thing COS needs from any of these is φ at $u = \omega_k$. Heavy tails ($c_4 > 0$) widen $[a, b]$ — exactly what keeps the short- T range from collapsing.

5 Cluster 5 — American options (L9)

Lecture 09 — Early exercise: optimal stopping and four pricers

Core idea. An optimal-stopping / free-boundary problem; price the Bermudan by backward recursion, then Richardson- extrapolate to the American.

- **Free boundary.** $v_0 = \sup_{\tau} \mathbb{E}^{\mathbb{Q}}[e^{-r\tau} \Lambda(S_{\tau}, \tau)]$; value $v = \max(\Lambda, \text{continuation})$; continuation region ($v > \Lambda$, BS PDE holds) vs stopping region ($v = \Lambda$), smooth pasting (v, v_S continuous) at the free boundary $S^*(t)$.
- **Call = European** on a non-dividend stock (dominating-strategy + parity bound $C \geq S - Ke^{-r(T-t)} > S - K$); the *put* breaks it because $Ke^{-r(T-t)} < K$ — banking the strike early to earn interest can win, so early exercise is genuinely optimal.
- **LCP / obstacle.** $(v - \Lambda) \geq 0$, $\mathcal{L}v \leq 0$, $(v - \Lambda)\mathcal{L}v = 0$ ($\mathcal{L}$ the BS operator); delta-hedging gives an equality for the European but only an inequality for the American (the holder controls exercise).

Bermudan backbone (log-space). $c(x, t_{n-1}) = e^{-r\Delta t} \mathbb{E}[v(X_{t_n}) | x]$, $v = \max(\Lambda, c)$. Four ways to compute c :

- **Binomial:** $V = \max(\Lambda, e^{-r\Delta t}[pV^+ + (1-p)V^-])$ — one line differs from European.
- **FD+LCP:** projected update $V = \max(\Lambda, \text{BTCS stencil})$ via **PSOR** (project *inside* each SOR sweep, not once at the end).
- **LSM:** regress discounted future cash flows on a basis (monomials / weighted Laguerre) over *in-the-money* paths; exercise if $\Lambda > \hat{c}$. Low-biased (suboptimal policy) with a competing in-sample foresight bias $\Rightarrow$ use out-of-sample. (8-path paper example: American 0.1144 vs European 0.0564.)
- **COS-Bermudan:** split $V_k(t_n) = G_k(a, x_n^*) + C_k(x_n^*, b, t_n)$ at the early-exercise point x_n^* (where $c = \Lambda$, found by Newton); G_k closed-form (χ, ψ) , C_k via FFT; cost $O(NK \log_2 K)$.

Reasoning line. All four pricers compute the same continuation value $\mathbb{E}[v(t_{n+1}) | x]$ differently; the COS range caveat bites *per step* because $c_2 = \sigma^2 \Delta t$ is small. Richardson extrapolates Bermudan $\rightarrow$ American in the number of exercise dates.

6 Cluster 6 — Exotics overview (L11)

Lecture 11 — Exotics map: payoff structure $\Rightarrow$ method

Core idea. A survey of path-dependent / multi-asset payoffs and the numerical method each one calls for.

- **Asian** (average-rate). Arithmetic average has no closed form (sum of correlated lognormals) $\Rightarrow$ MC with the *geometric*-average control variate (geometric Asian *is* lognormal: $\sigma_G^2 \rightarrow \sigma^2 T/3$, Black-style closed form). Under stochastic vol, the Fouque–Han 3-D PDE reduces to 1-D by the Vecer numeraire/homogeneity change of variable.
- **Digital / binary.** Price = $e^{-rT} N(d_2)$ ($= e^{-rT} \mathbb{Q}(S_T > K)$); $\partial C / \partial K$ (Breen–Litzenberger) and a centred call-spread replicate it. Discontinuous payoff $\Rightarrow$ delta / pin-risk blow-up ($\sim 1/\varepsilon$ notional) near K at expiry — the recurring hedging trap.
- **Basket / worst-of.** Correlation-driven; MC dominates, or moment-matching / 2-D COS (factorises when $\rho = 0$). No clean COS for ≥ 4 -asset baskets.
- **Cliquet.** Forward-starting capped/floored periodic returns; sensitive to the forward skew.
- **Barrier.** See L10.

Reasoning line. Method follows structure: averaging / multi-asset / high dimension $\Rightarrow$ Monte Carlo; smooth, low-dimensional with a known cf $\Rightarrow$ COS; the digital's discontinuity is the recurring

hedging difficulty.

7 Quick-reference cards

COS in five lines (say it cold)

1. Price = $e^{-rT} \int_a^b V(y) f(y | x_0) dy$; I know φ , not f .
2. Truncate to $[a, b] = c_1 \pm L\sqrt{c_2 + \sqrt{c_4}}$ and cosine-expand f there.
3. Density coefficients come straight from the cf: $A_k = \frac{2}{b-a} \operatorname{Re}\{\varphi(\omega_k) e^{-i\omega_k a}\}$.
4. Payoff coefficients V_k are closed-form (χ_k, ψ_k) .
5. Pair them: $P \approx e^{-rT} \sum_k' \operatorname{Re}\{\varphi e^{-i\omega_k a}\} V_k - O(N)$, exponential convergence; the only floors are $[a, b]$ and N .

Convergence rates at a glance

Method	Error	Note
Monte Carlo	$O(N^{-1/2})$	dimension-free; the benchmark
QMC	$\sim (\log N)^d / N$	needs low effective dimension
Binomial / FTCS	$O(1/M), O(k + h^2)$	FTCS needs $k \leq h^2/2$
Crank–Nicolson	$O(k^2 + h^2)$	implicit, unconditionally stable
COS (smooth f)	exponential in N	floors at range truncation
COS (finite smoothness)	algebraic $O(k^{-(\beta+1)})$	e.g. VG small ν

Method-selection table

Problem	Preferred method	Rate
European, cf known (BS/Heston/Lévy)	COS	exponential in N
European, density known (GBM)	Black–Scholes closed form	exact
Low-dim PDE, vanilla/American	Finite differences (CN; PSOR for LCP)	$O(k^2 + h^2)$
American/Bermudan, cf known	COS-Bermudan (Newton + FFT)	exponential in N
American, general / high-dim	LSM regression Monte Carlo	$O(N^{-1/2})$
Discrete barrier, cf known	COS backward recursion (FFT)	$O((M-1)N \log N)$
High-dim / basket / arithmetic Asian	Monte Carlo (+ variance reduction / QMC)	$O(N^{-1/2})$

Characteristic functions to keep at hand

Model	Characteristic function (argument u , horizon t)
Black–Scholes	$\exp(iu(r - \frac{1}{2}\sigma^2)t - \frac{1}{2}\sigma^2 u^2 t)$
Heston	$\exp(\alpha(t, u) + \beta(t, u)v_0 + iu \ln S_0)$, α, β from Riccati ODEs
Merton jumps	$\exp(iu\mu t - \frac{1}{2}\sigma^2 u^2 t + \lambda t(e^{iu\mu_J - \frac{1}{2}\sigma_J^2 u^2} - 1))$
Kou jumps	diffusion $\times \exp(\lambda t(\frac{p\eta_1}{\eta_1 - iu} + \frac{(1-p)\eta_2}{\eta_2 + iu} - 1))$
Variance-Gamma	$(1 - iu\theta\nu + \frac{1}{2}\sigma^2 \nu u^2)^{-t/\nu}$

The cf is the only model-specific input COS needs — the V_k and the cosine sum are identical across models.

Numbers & facts worth memorising

- $(dW)^2 = dt$; $\mathbb{E}[e^{\sigma W_t}] = e^{\sigma^2 t/2}$; $\ln(S_T/S_0) \sim N((r - \frac{1}{2}\sigma^2)T, \sigma^2 T)$ under $\mathbb{Q}$.
- BS cumulants: $c_1 = \ln \frac{S_0}{K} + (r - \frac{1}{2}\sigma^2)T$, $c_2 = \sigma^2 T$, $c_3 = c_4 = 0$; range $= c_1 \pm L\sigma\sqrt{T}$.
- $N(d_2) = \mathbb{Q}(S_T > K)$; $\Delta_{\text{call}} = N(d_1)$; Vega $= S\sqrt{T}\phi(d_1)$.
- Feller $2\kappa\bar{v} \geq \eta^2$; CFL $k \leq h^2/2$; COS frequency $\omega_k = k\pi/(b-a)$.
- Costs: European COS $O(N)$; discrete barrier $O((M-1)N \log_2 N)$; Bermudan COS $O(NK \log_2 K)$; MC $O(N^{-1/2})$.

The through-line, once more

Discounted expectation $\rightarrow$ if density known, closed form / PDE / tree; if only cf known, **COS** (cosine coefficients of f from φ , paired against closed-form payoff coefficients V_k , summed in $O(N)$). Early exercise and discrete barriers reuse COS as a *backward recursion*; high-dimensional and averaging payoffs fall back to *Monte Carlo*. Always set $[a, b] = c_1 \pm L\sqrt{c_2 + \sqrt{c_4}}$ and watch it collapse at short maturity — the one trap Fang is most likely to probe.

8 Derivation skeletons — the logical chain, not the algebra

For each marquee result, the *steps* (so you can rebuild the algebra live, but recall the route under pressure). Full algebra is in the lecture notes.

BS PDE (delta-hedging).

Form $\Pi = V - \Delta S \rightarrow$ Itô on $V \rightarrow$ choose $\Delta = V_S$ to cancel the dW term $\rightarrow$ the residual is riskless so it must earn r ($d\Pi = r\Pi dt$) $\rightarrow$ collect terms $\Rightarrow V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0$; μ never appears.

Closed-form call.

$C = e^{-rT} \mathbb{E}^\mathbb{Q}[(S_T - K)^+] \rightarrow$ split into $\mathbb{E}[S_T \mathbf{1}] - K\mathbb{E}[\mathbf{1}] \rightarrow$ the S_T term is handled by a measure change (share numeraire) giving $N(d_1)$, the indicator gives $\mathbb{Q}(S_T > K) = N(d_2) \rightarrow$ complete the square in the lognormal integral for the $d_2 \rightarrow d_1$ shift.

COS coefficient A_k .

Start from $A_k = \frac{2}{b-a} \int_a^b f \cos(\omega_k(y-a)) dy \rightarrow$ write $\cos = \text{Re}\{e^{i\cdot}\}$, pull out $e^{-i\omega_k a} \rightarrow$ extend $\int_a^b \rightarrow \int_{\mathbb{R}}$ (the one approximation) $\rightarrow$ the remaining integral is $\varphi(\omega_k) \Rightarrow A_k \approx \frac{2}{b-a} \text{Re}\{\varphi(\omega_k) e^{-i\omega_k a}\}$.

COS price.

Insert $f \approx \sum' A_k \cos(\cdot)$ into $e^{-rT} \int V f \rightarrow$ swap $\sum / \int$ (uniform convergence) $\rightarrow$ define $V_k \rightarrow P \approx e^{-rT} \frac{b-a}{2} \sum' A_k V_k \rightarrow$ substitute $\frac{b-a}{2} A_k = \text{Re}\{\varphi e^{-i\omega_k a}\}$.

Cumulant range.

Need $[a, b]$ to hold (almost) all mass of $X_T \rightarrow$ centre at the mean c_1 , scale by $\sqrt{c_2}$ (an “ L -sigma” box), fatten by $\sqrt{c_4}$ for heavy tails $\Rightarrow c_1 \pm L\sqrt{c_2 + \sqrt{c_4}}$.

Reflection principle.

$\{\Xi \geq a\} = \{\tau_a \leq T\} \rightarrow$ paths past a finish above/below a with prob $\frac{1}{2} \rightarrow \mathbb{P}(\tau_a \leq T) = 2\mathbb{P}(W_T \geq a) \rightarrow$ differentiate for p_Ξ ; reflect the endpoint across a ($b \mapsto 2a - b$) for the joint density.

Heston cf.

Affine in $(\ln S, v) \rightarrow$ ansatz $\varphi = \exp(\alpha + \beta v_0 + iu \ln S_0) \rightarrow$ substitute into the Feynman–Kac PDE $\rightarrow$ match constant and v -linear terms $\Rightarrow$ a Riccati ODE for β and a quadrature for $\alpha \rightarrow$ solve β via $g = 1/\beta$, integrate α .

Discrete-barrier COS.

Survival = $\prod_m \mathbf{1}_{\{x_m < h\}}$ $\rightarrow$ Markov factorises the joint density $\rightarrow$ nested 1-D integrals = backward recursion $\rightarrow$ each step is a European-COS continuation $\rightarrow$ knock-out splits V_k at $h \rightarrow$ the coupling is Toeplitz+Hankel $\rightarrow$ FFT.

9 One question per lecture to be airtight on

- L1. Why do the physical probabilities never enter the replication price?
 - L2. Pinpoint where $\Delta = V_S$ is forced and where μ cancels in the BS derivation.
 - L3. Prove the implied vol is unique; why is Newton unsafe deep OTM?
 - L4. Derive the Milstein term; state strong vs weak orders.
 - L5. Derive the FTCS stability bound $\nu \leq \frac{1}{2}$ by von Neumann; why is BTCS unconditional?
 - L6. Derive the European COS formula and say exactly where the cf enters.
 - L7. Write the Heston cf and state + justify the Feller condition.
 - L8. Derive the compound-Poisson cf and the risk-neutral drift correction.
 - L9. Prove American call = European; why does the put differ?
 - L10. Why is discrete-barrier COS a backward recursion, and why does the FFT reappear?
 - L11. Give the digital price $e^{-rT} N(d_2)$ and explain its hedging blow-up near expiry.
- COS. How do you pick $[a, b]$, what breaks at short maturity, and how did you fix it in Assignment 2?